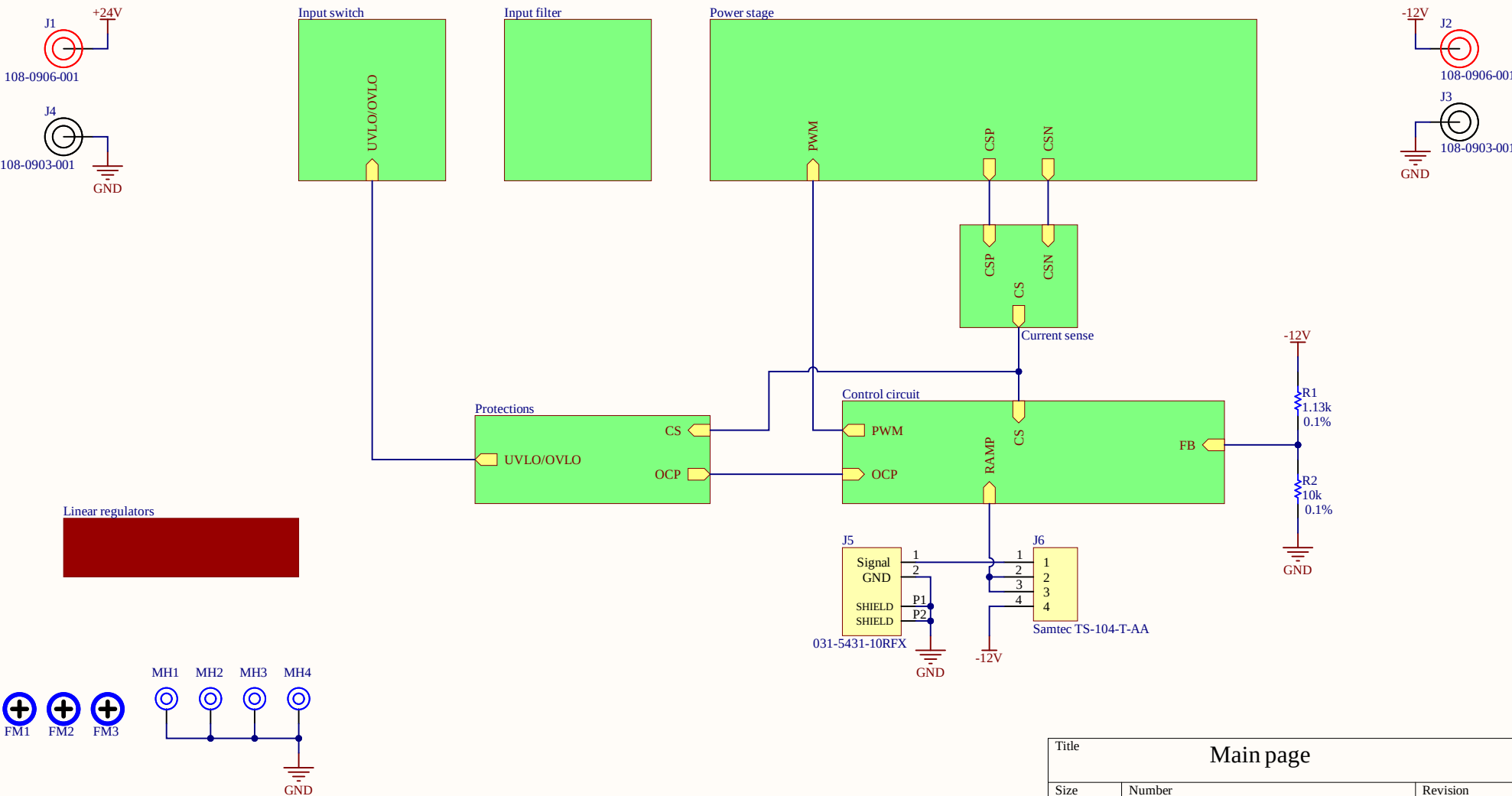
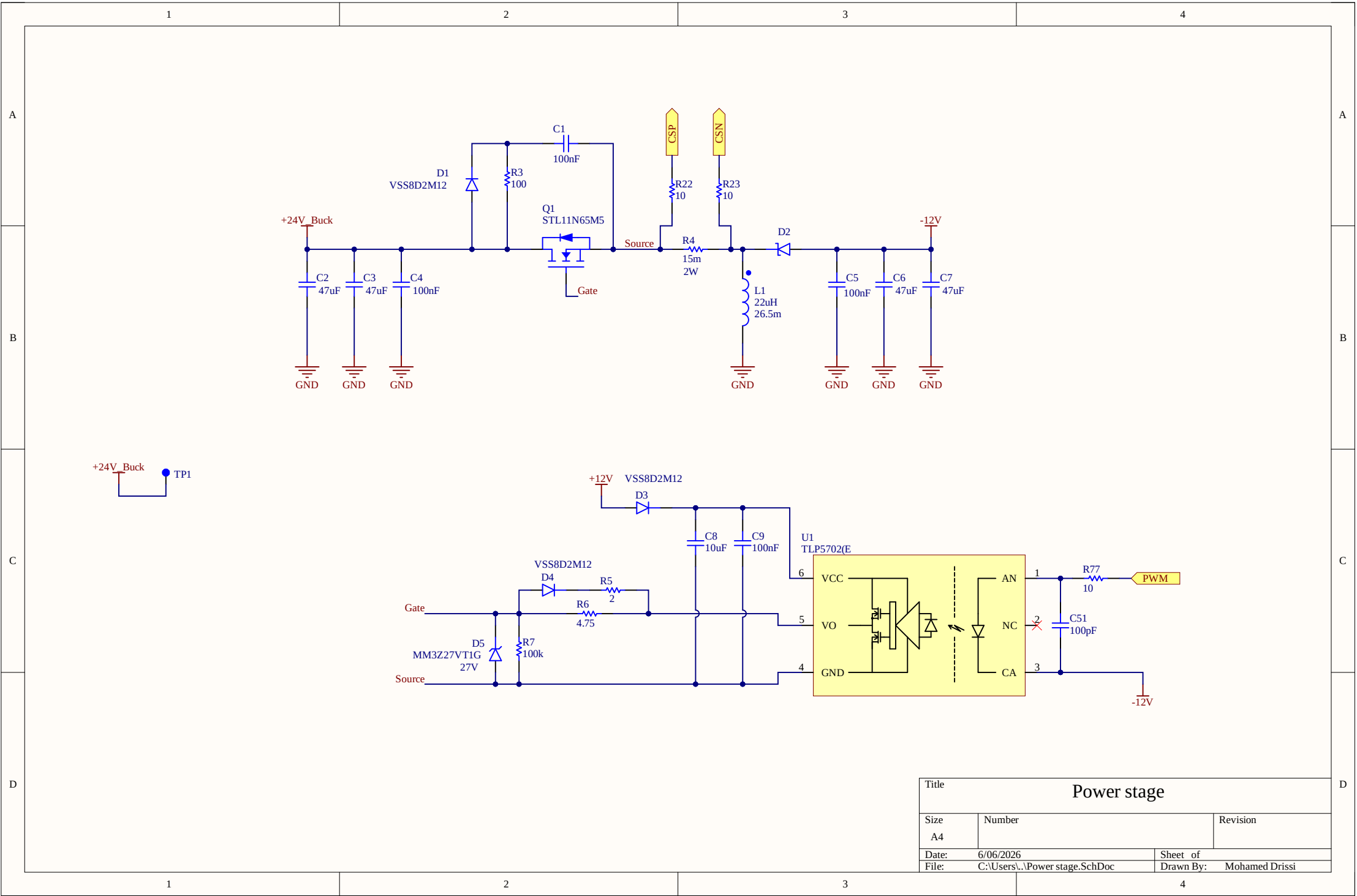
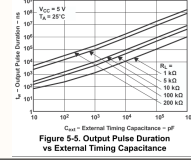
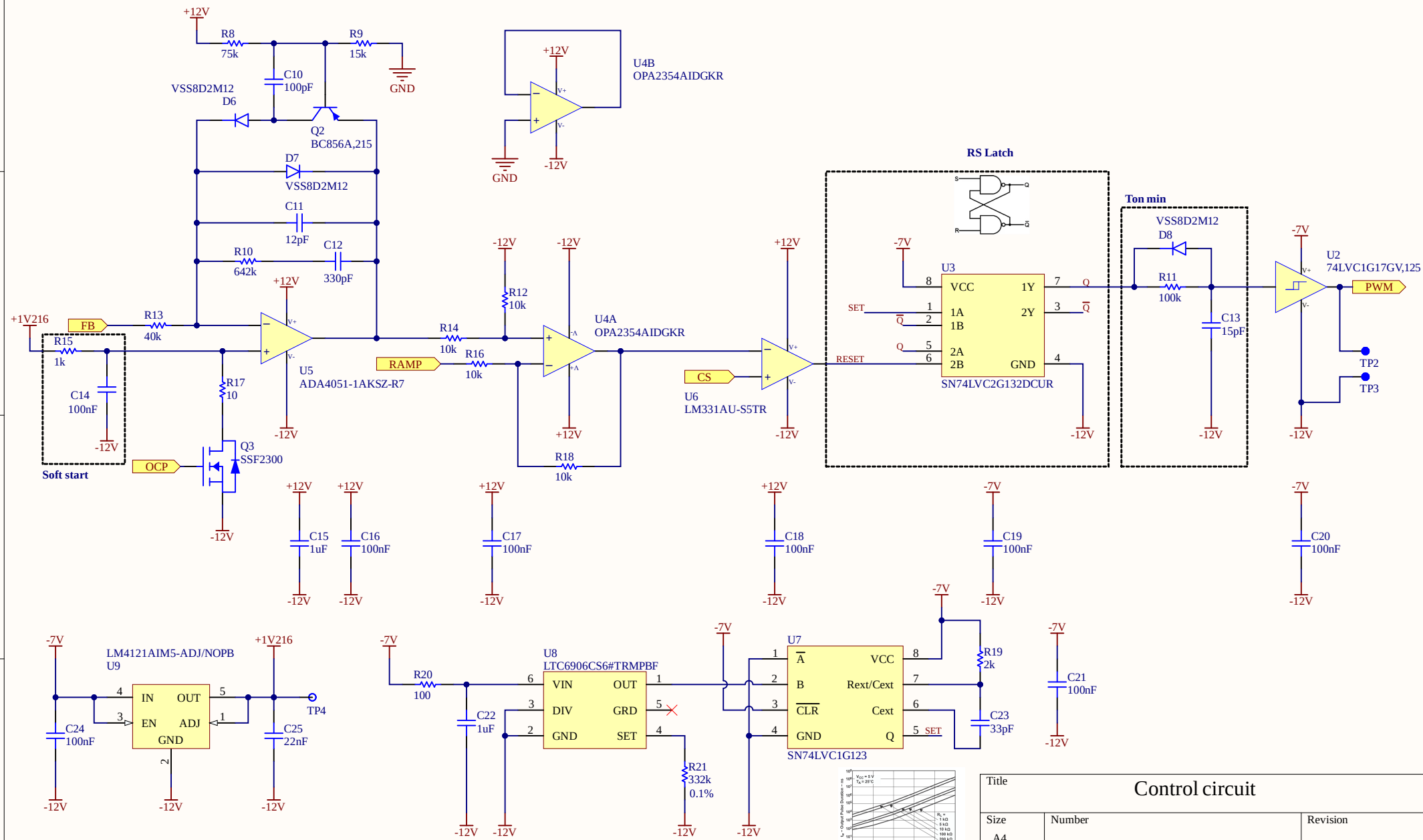


Inverting Buck-Boost

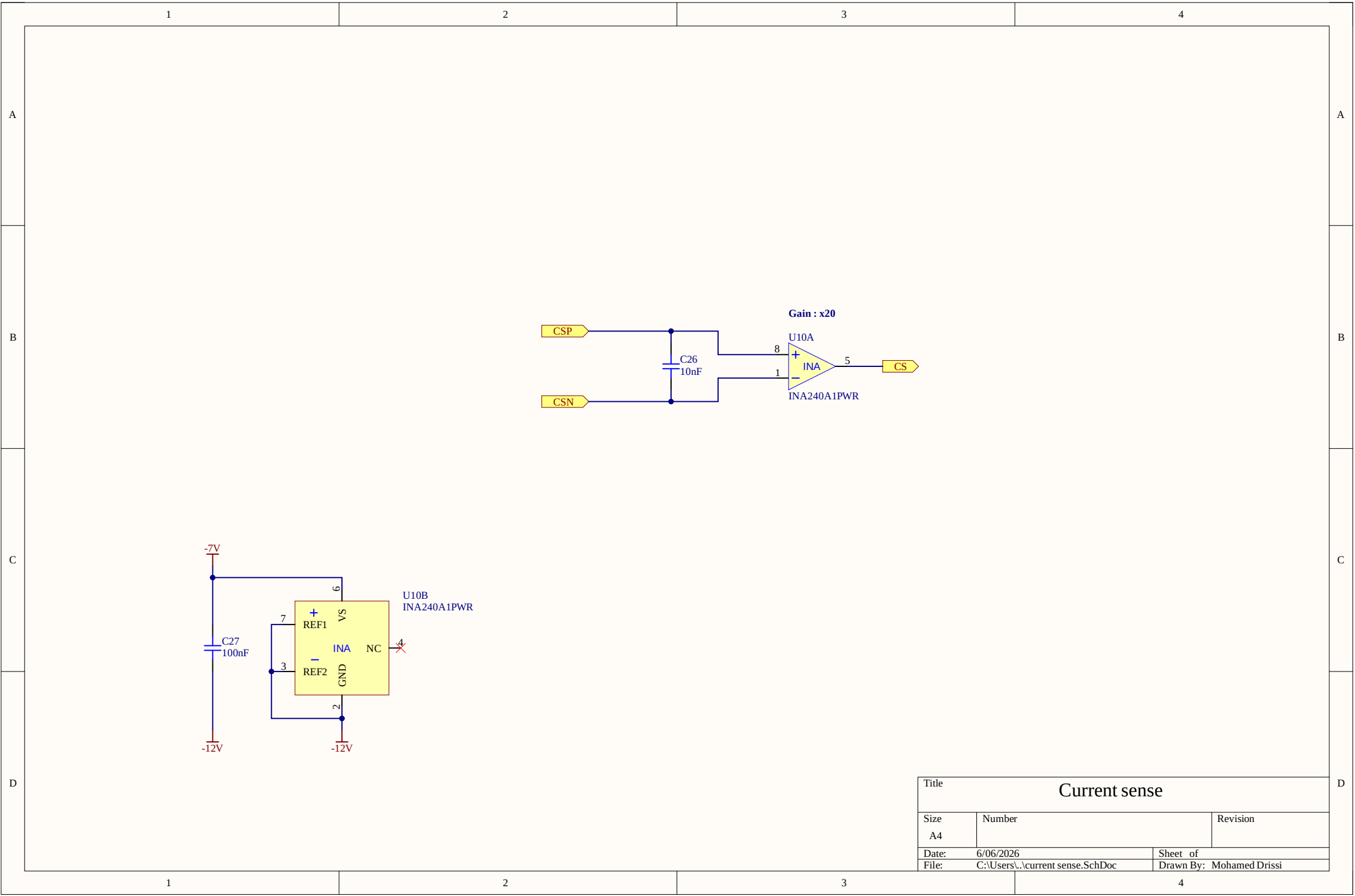


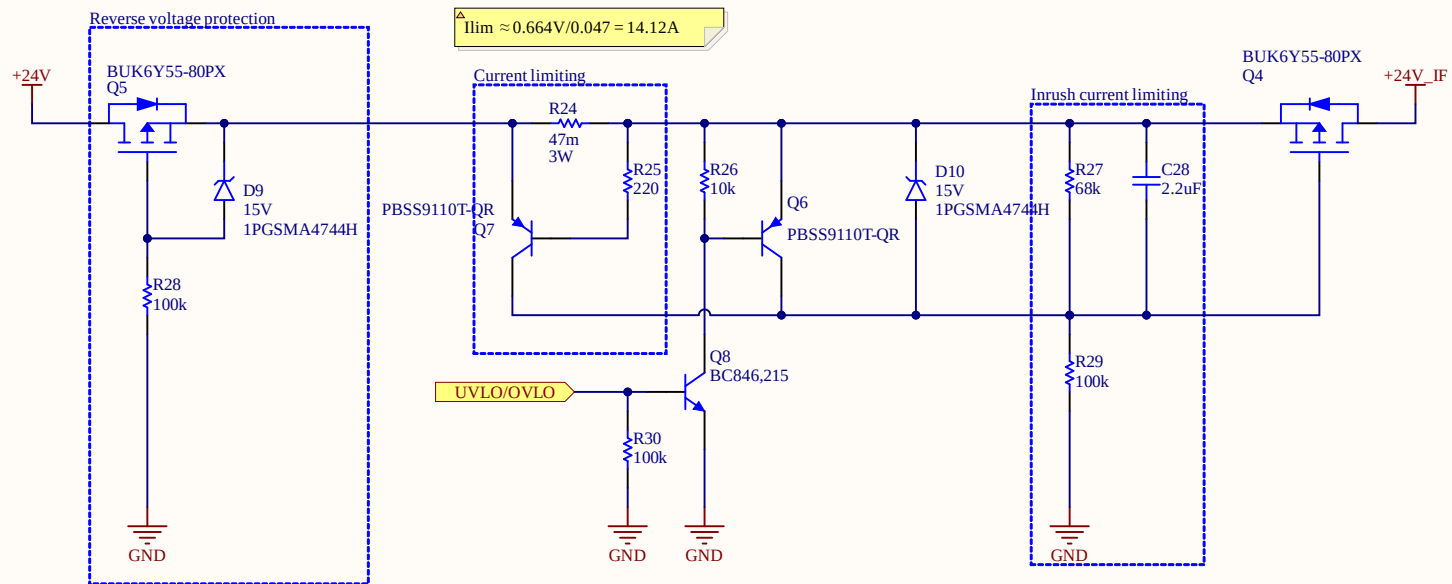
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Size	Number	Revision
A4		
Date:	6/06/2026	Sheet of
File:	C:\Users\Main.SchDoc	Drawn By: Mohamed Drissi



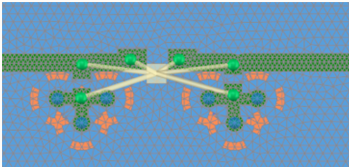
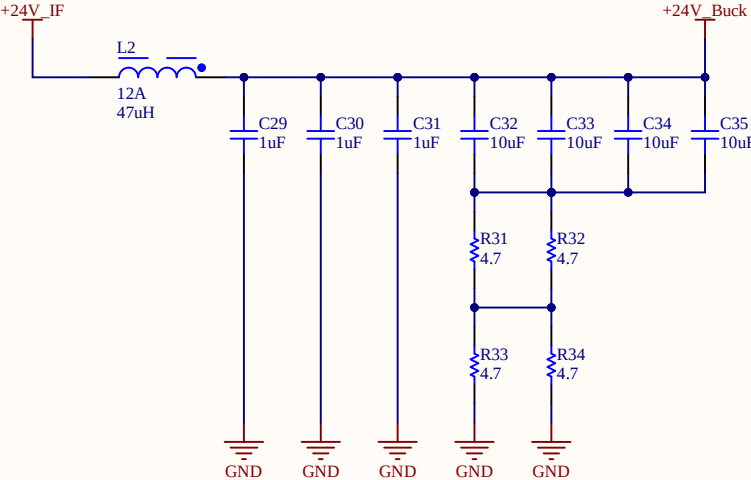


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Size	Number	Revision
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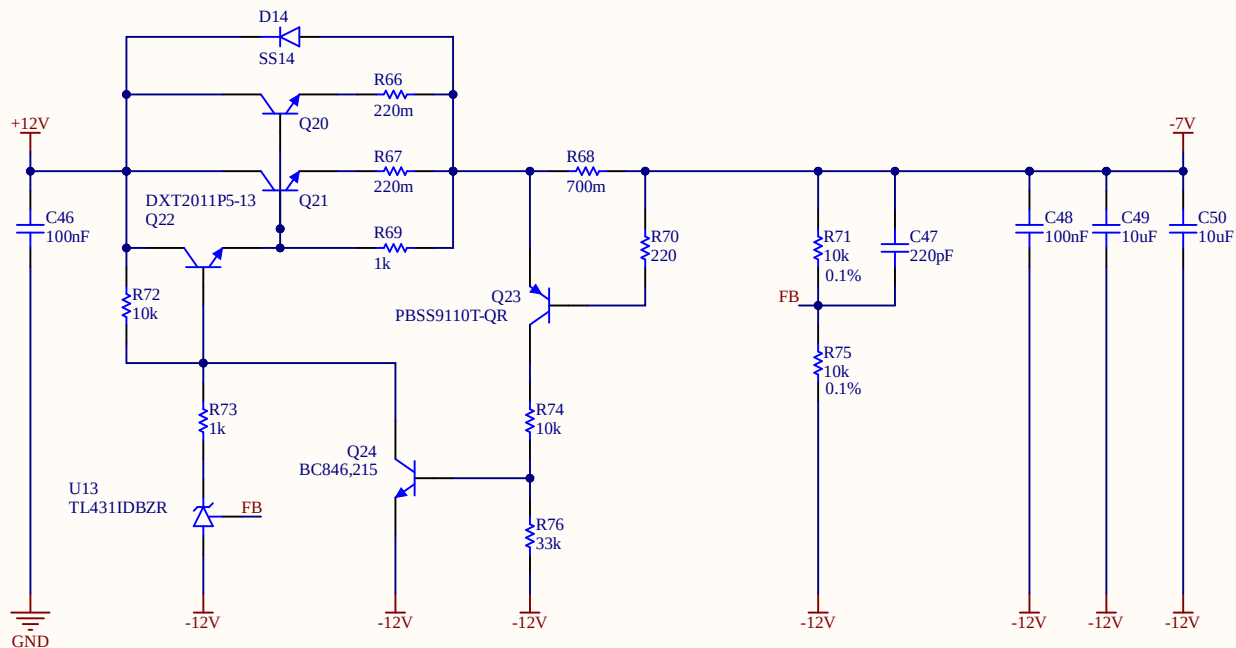
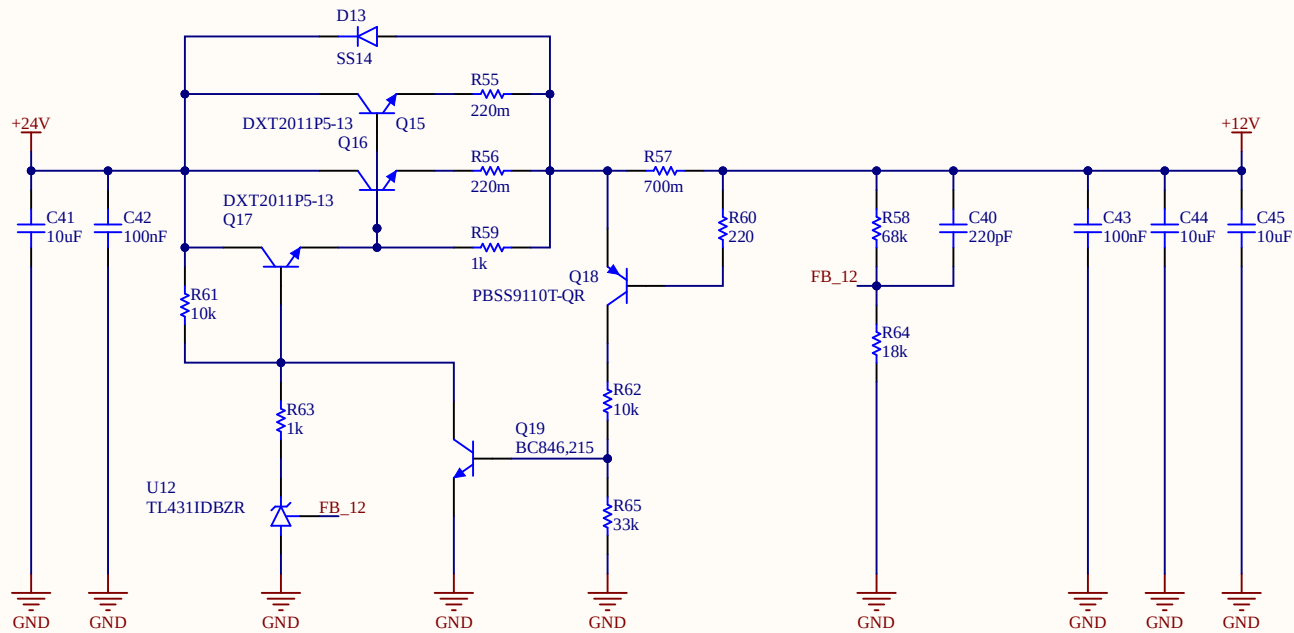




Title		
Input switch		
Size	Number	Revision
A4		
Date:	6/06/2026	Sheet of
File:	C:\Users\...\Input switch.SchDoc	Drawn By: Mohamed Drissi



Title			Input filter	
Size	Number		Revision	
A4				
Date:	6/06/2026		Sheet of	
File:	C:\Users\...\Input filter.SchDoc		Drawn By: Mohamed Drissi	



The NPN transistors can dissipate 3W if the device mounted with the exposed collector pad on 50mm x 50mm 2oz copper that is on a single-sided 1.6mm FR4 PCB

Title		
Linear regulators		
Size	Number	Revision
A4		
Date:	6/06/2026	Sheet of
File:	C:\Users\Linear regulators.SchDoc	Drawn By: Mohamed Drissi

Line#	Name	Description	Designator	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Stock# 1
1	1PG8M414H1	Zener Diode, 15V,0.5W, 5%, 10W, Silicon, Unidirectional, DO-214AC	CR, CR0	2							
2	031-5431-100FX	RF IEC Connector, Female, Panel Mount, Board And Panel Mount, Solder Terminal, Jack	J5	1							
3	74LVC1G170V,126	Schmitt Trigger Buffer, Single, 1 Input, 5 Pins, TSSOP, 74LVC126/17	U2	1							
4	108-0903-001	Reverse Bias Connector, 4mm, 40A, Panel Mount, 15-A, 7-Pin, Tin/Plated Contacts, Back	J1,J4	2							
5	108-0906-001	Right Angle Dual Port D-Sub Connector	J, J2	2							
6	5000	Clear/Violet, 1/2 Pin, Polyethylene, Natural, 65FR, (Snap Series), IE Connectivity 500000-1	TP1, TP2, TP3, TP4	4							
7	7443634700	Power Diode Shottky Viewround 47uA 15% 100Hz Ferrite 12A 0.25A300mV 0.25A300mV	L2	1							
8	885012000026	WCAP-C2KP0603 (600MWeic) Ceramic Capacitor 0.063uF 0.05W 1.6V	C12	1							
9	885012200073	Cap, 10k, 50V, Mfc., 1250 Parts Compliance: Yes Mfr's Bolebrook 885012200073	CR, CR2, CR3, CR4, CR5, CR6, CR7, CR8, CR9	10							
10	885012200005	WCAP-C2B1206 (600MWeic) ceramic capacitor 0.12uF 0.05W 1.6V	CR18, CR9, CR4, CR1, CR7	5							
11	885012200003	WCAP-C2T1210 (600MWeic) Ceramic Capacitor 0.12uF 0.05W 1.6V	CR18	1							
12	CP40651-148200T	Op Amp Single Matchlower Amplifier REEVC55V 5Pin 16: 70 W 18	J48	1							
13	AT0603R00140N	100mW Thin Film Resistor ±25ppm/°C, ±0.1% 400/0003 Chip Resistor - Surface Mount RCH5	R63	1							
14	RC346-235	RC346 Series 65 V 100 mA SMT NPN General Purpose Transistor - SC7-23	QR, QR3, QR4, QR8, QR4	5							
15	RC356A-235	RC356 Series 65 V 100 mA SMT PNP General Purpose Transistor - SC7-23	QR, QR, QR1, QR2	2							
16	BL40752-00P5	50V, 4-Pin, Surface Mount, M2025	CR4, CR2	2							
17	RC0804-45M6-CX	Diode Zener 5.0V 200mW SC1023	CR12	3							
18	C0603C220R9AC1U	MLCC Ceramic Capacitors MLC - SMD SMT 220pF 50V 0.05W 10% - C0603C220R9AC1U	CR5, CR6, CR2	3							
19	C0603C220R9AC1U	MLCC Ceramic Capacitors MLC - SMD SMT, REVERSE: C0603C220R9AC1U 22 pF 0603, 50 V, 10 V	CR9	1							
20	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR4, CR7	2							
21	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR6	1							
22	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
23	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
24	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
25	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
26	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
27	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
28	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
29	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
30	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
31	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
32	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
33	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
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35	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
36	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
37	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
38	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
39	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
40	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
41	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
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57	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
58	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
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74	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
75	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
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96	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
97	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
98	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
99	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							
100	C0603C220R9AC1U	10V 220pF C0603C220R9AC1U MLCC Ceramic Capacitors MLC - SMD SMT RCH-5	CR3	1							